

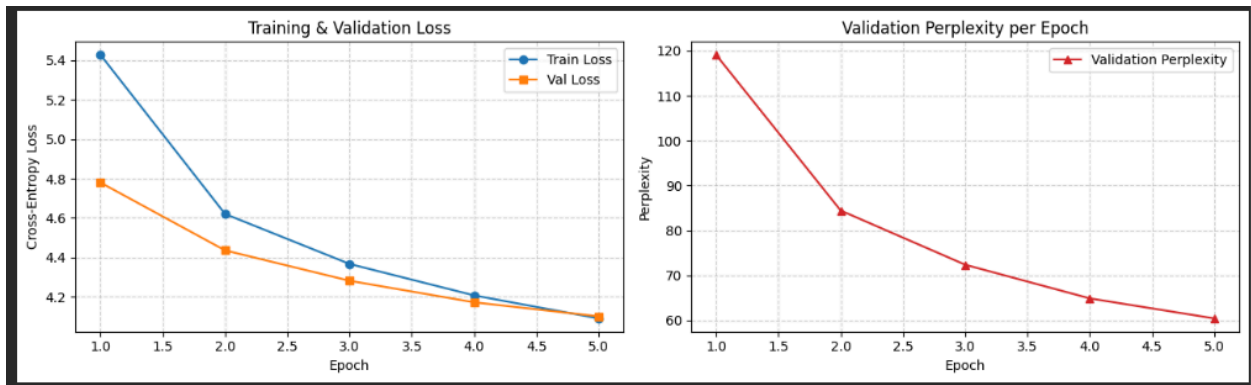
TASK 2 REPORT

Task 2.2 GPT losses:

```
Epoch 1/5 | Train Loss: 5.4276 | Val Loss: 4.7798 | Val Perplexity: 119.08  
Saved best model to saved_model/causal_transformer_best.pt  
Epoch 2/5 | Train Loss: 4.6207 | Val Loss: 4.4360 | Val Perplexity: 84.44  
Saved best model to saved_model/causal_transformer_best.pt  
Epoch 3/5 | Train Loss: 4.3672 | Val Loss: 4.2818 | Val Perplexity: 72.37  
Saved best model to saved_model/causal_transformer_best.pt  
Epoch 4/5 | Train Loss: 4.2075 | Val Loss: 4.1726 | Val Perplexity: 64.88  
Saved best model to saved_model/causal_transformer_best.pt  
Epoch 5/5 | Train Loss: 4.0908 | Val Loss: 4.1013 | Val Perplexity: 60.42  
Saved best model to saved_model/causal_transformer_best.pt  
Best epoch: 5  
Best validation loss: 4.1013  
Best validation perplexity: 60.42
```

Test loss:

```
test loss: 4.1073 | test perp: 60.78
```



Sample test:

```
46  
47  
48 print(generate_text("The history of science", max_new_tokens=30, temperature=0.8))
```

The history of science is not a collection of information. It was discovered in 1843 by James R. Johnson. It is now in the National Museum of Science and Sciences

Heatmap:

